J. Additional results on induction learning

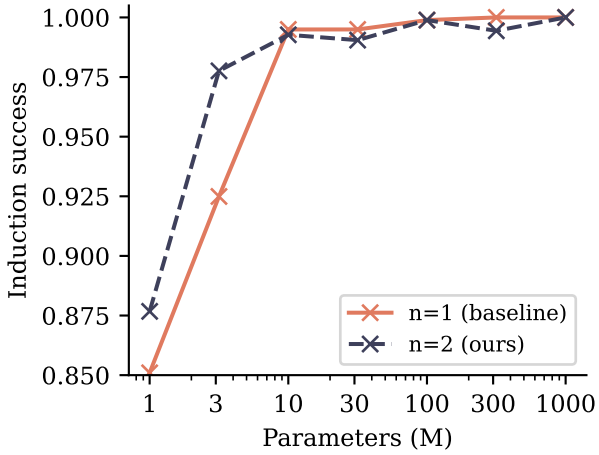


Figure S14: **Induction capability of n-token prediction models trained on higher-quality data.** Shown is accuracy on the second token of two token names that have already been mentioned previously. Training on a 9:1 mix of a books dataset and the children storiy dataset, we observe that induction capability forms significantly earlier in training (not shown here) and to a higher degree. We believe that this is explained both because our evaluation dataset no longer contains out-of-distribution tokens (Section 4.1) and because the higher-quality data contained in the books dataset makes induction necessary earlier on (especially for small models, cf. Singh et al. (2023)). In particular, by enforcing the formation of induction capability in the model by means of the dataset – instead of the loss – the advantage of 2-token prediction models on this task disappears except for the smallest models: feature learning converts the task into a pure next-token prediction task.

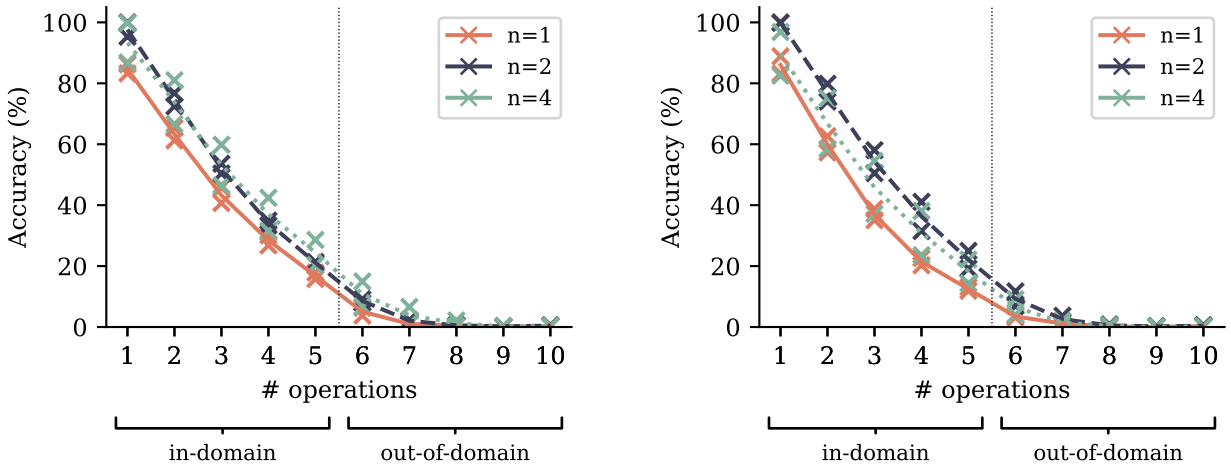
K. Additional results on algorithmic reasoning

We investigate the following *computation-sharing hypothesis* for explaining the efficacy of multi-token prediction as training loss.

> The prediction difficulty of different tokens in natural text varies greatly. Some tokens may be the continuations of partial words that are uniquely determined from their preceding context without any effort, while others may require to predict theorem names in difficult mathematical proofs or the correct answer to an exam question. Language models with residual connections have been shown to refine their output token distribution with each successive layer, and can be trained with early exit strategies that spend variable amounts of computational resources per token position. Multi-token prediction losses explicitly encourage information-sharing between adjacent token positions and can thus be viewed as a method to learn allocating computational resources in language models more efficiently to the tokens that benefit most of it.

To check the truth of this hypothesis, we augment the polynomial arithmetic task from Section 4.2 with a varying number of *pause tokens* (Goyal et al., 2023) inserted between the question and a token that denotes the beginning of the answer. Pause tokens introduce additional computational resources that can be expended for computations that are expected to be useful later on in the sequence, in other words: to start thinking about the answer. According to the *computation-sharing hypothesis*, multi-token prediction models learn information-sharing and thus computation-sharing between token positions more easily, and may be better at making use of these additional computational resources than next-token prediction models are. In Figure S15, we show the evaluation results on the polynomial arithmetic task with a fixed number of pause tokens inserted both at training and evaluation time. Multi-token prediction models likewise outperform next-token prediction models on these task variants across task difficulties and model sizes. However, we do not see strong evidence of a widening or shrinking of this gap i.e. we cannot conclude from these experiments on the veracity of the computation-sharing hypothesis.

In Table S11, we report results from another experiment in the same spirit: by adding spaces and newlines to HumanEval and MBPP prompts, we add "pause tokens" in a somewhat natural way. According to these results, multi-token prediction models have a slight advantage at using this additionally provided compute, but the effect is marginal.



(a) 5 pause tokens

(b) 10 pause tokens

Figure S15: **Accuracy on a polynomial arithmetic task with varying number of operations per expression and pause tokens.** We train and evaluate models on the polynomial arithmetic task described in Section 4.2, modified by the addition of *pause tokens* (Goyal et al., 2023): between the question and the equality sign that indicates the beginning of the answer, we add a constant number of pause tokens both in training and evaluation. For both a variant with five and with ten pause tokens, respectively, we observe comparable improvements from using multi-token prediction to the ones obtained in the case without pause tokens (Figure 8).

Table S11: **Utilization of additional whitespace tokens in code benchmarks.**

Task	Whitespace	$n = 1$	$n = 4$
APPS/Intro	spaces + newline	+0.21	**+0.34**
APPS/Intro	newline	**+0.79**	+0.69
HumanEval	spaces + newline	-0.72	**-0.16**
HumanEval	newline	-0.26	**+0.10**
MBPP	spaces + newline	-0.10	**-0.06**
MBPP	newline	**+0.03**	-0.08
Average		-0.01	**+0.14**

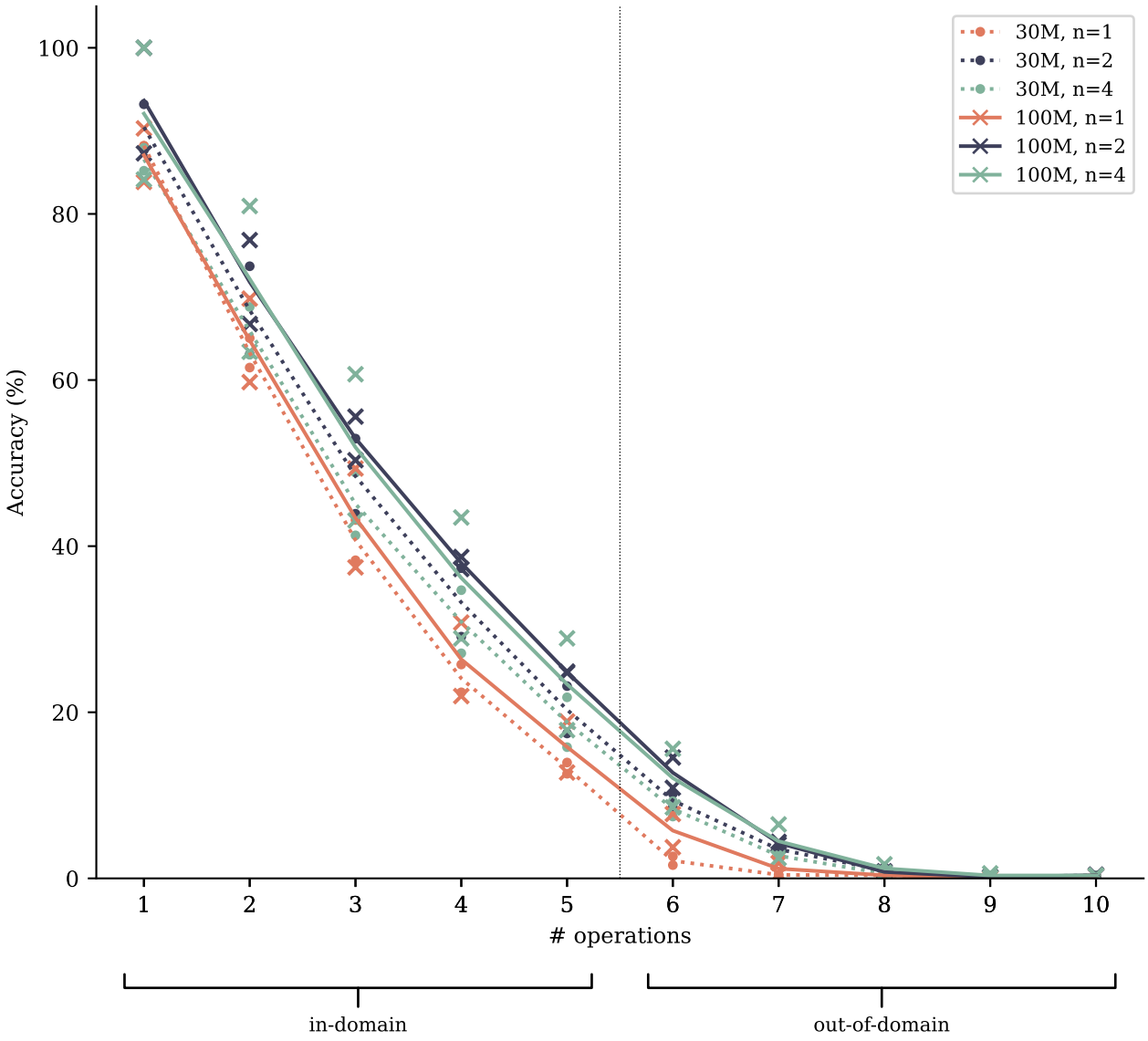


Figure S16: **Accuracy on a polynomial arithmetic task for two model sizes.** We train and evaluate models with 30M and 100M parameters on the polynomial arithmetic task described in Section 4.2. Tripling the model size has a smaller effect on performance than replacing next-token prediction loss by multi-token prediction. Shown are two independent runs per configuration and their means, the 100M parameter models being identical to the ones in Figure 8.

Table S12: **Optimal temperatures for all numbers in table 1**

Training data	Vocabulary	n	MBPP			HumanEval			APPS/Intro		
			@1	@10	@100	@1	@10	@100	@1	@10	@100
313B bytes (0.5 epochs)	bytes	1	0.2	0.8	0.8	0.1	0.8	0.8	0.8	0.8	0.8
		8	0.1	0.8	0.8	0.1	0.8	0.8	0.4	0.4	0.4
		16	0.1	0.8	0.8	0.1	0.8	0.8	0.4	0.4	0.4
		32	0.1	0.4	0.8	0.1	0.4	0.8	0.1	0.4	0.4
200B tokens (0.8 epochs)	32k tokens	1	0.1	0.8	0.8	0.1	0.8	0.8	0.1	0.4	0.8
		2	0.1	0.8	0.8	0.2	0.8	0.8	0.4	0.4	0.8
		4	0.1	0.8	0.8	0.1	0.8	0.8	0.2	0.8	0.8
		6	0.1	0.8	0.8	0.2	0.8	0.8	0.4	0.4	0.8
		8	0.1	0.8	0.8	0.1	0.8	0.8	0.2	0.4	0.8
1T tokens (4 epochs)	32k tokens	1	0.1	0.8	0.8	0.1	0.8	0.8	0.1	0.4	0.8
		4	0.1	0.8	0.8	0.2	0.8	0.8	0.4	0.8	0.8

L. Additional intuitions on multi-token prediction

L.1. Comparison to scheduled sampling

In Section 5.2, we argued that multi-token prediction reduces the distribution mismatch between teacher-forced training and autoregressive evaluation of language models. Scheduled sampling (Bengio et al., 2015) is a curriculum learning method that likewise aims to bridge this gap in sequence prediction tasks by gradually replacing more and more input tokens with model-generated ones.

While effective in areas such as time series forecasting, scheduled sampling is, in our opinion, inapplicable to language modelling due to the discrete nature of text. Replacing ground truth input sequences by interleavings of ground truth and model-generated tokens frequently results in ungrammatical, factually wrong or otherwise incoherent text, which should be avoided at all cost. Moreover, unlike multi-token prediction, the technique originally developed for recurrent neural networks cannot easily be adapted for parallel training setups like the ones of transformer models.

L.2. Information-theoretic argument

We give details on the information-theoretic terms appearing in the decomposition in Section 5.2 and derive a relative version that similarly allows to decompose multi-token prediction losses. As in Section 5.2, denote by X the next token and by Y the second-next one, and omit conditioning on the preceding context C for ease of notation. In Section 5.2, we decomposed $H(X) + H(Y)$—the quantity of interest for 2-token prediction models—as follows:

$$H(X) + H(Y) = H(X \mid Y) + 2I(X;Y) + H(Y \mid X). \tag{3}$$

Let us explain each of the terms. The entropy terms denote the uncertainty contained in the ground-truth random variables X and Y. [2] The term $H(Y \mid X)$ is a classical next-token entropy for the prefix (C, X). The conditional entropy $H(X \mid Y)$ is a more theoretical entity not modelled by causal models. It describes the uncertainty about X given the prefix C and suffix Y, and therefore captures the local variations of X that do not affect the continuation of the text Y. The mutual information $I(X;Y)$ on the other hand describes the information about Y contained in X (and vice versa) and therefore captures the variations of X which constrain the continuation of the text.

However, the argument given in Section 5.2 relies on the assumption that multi-token prediction losses obey a similar decomposition as the sum of the ground-truth entropies themselves. Let us make this rigorous. Denote by $p(x, y)$ the joint distribution of X and Y, by $p(x)$ (short for $p_X(x)$) the marginal distribution of X and by $p(y)$ the one of Y. Denote the densities of the model's predictions by $q(x, y)$, $q(x)$ and $q(y)$, respectively, conditional distributions by $p(x \mid y)$ and Kullback-Leibler divergence from q to p by $D(p \parallel q)$ and cross-entropy from q to p by $H(p, q)$.

Definition L.1. The *conditional cross-entropy* $H(p_{X|Y}, q_{X|Y})$ of X conditioned on Y from q to p is defined as the

[2] In particular, they do not refer to *model* predictions.